Review of Robotics Fundamentals

Robotics is the art, knowledge base, and know-how of designing, applying, and using robots in human endeavors. These are compelling elements of today's industry. It may be used in manufacturing environments, underwater and space exploration, researching human and animal behavior, aiding disabled people, for transportation and delivery, for military purposes, or even for fun.

- *ISO 8373* defines a **robot** as ***"an automatically controlled, reprogrammable, multipurpose manipulator with three or more axes."***
- The *Robot Institute of America* designates a **robot** as ***"a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of various tasks."***
- The definition offered by *Merriam-Webster Dictionary* states that a **robot** is ***"a machine that looks like a human being and performs various complex acts (as walking or talking) of a human being."***

Robot History

Karel Čapek wrote the play **Rossum's Universal Robots,** in which the theme is one of futuristic man-made workers created to automate the work of humans, thus alleviating their burden.

- As Čapek wrote his play, he turned to his older brother for a name to call these beings, which coined the word **"robot."**
- The Czech word ***"Robotnik"*** refers to **a peasant or serf**, while ***"robota"*** means **drudgery or servitude**.

Isaac Asimov wrote a series of short stories that involved robot themes. In his short stories, Asimov introduced what would become the ***"Three Laws of Robotics."***

- **Zeroth Law**: A robot may not injure humanity or, through inaction, allow humanity to come to harm.
- **First Law**: A robot may not injure a human being or, through inaction, allow a human being to come to harm unless this would violate a higher-order law.
- **Second Law**: A robot must obey the orders given to it by human beings, except where such orders would conflict with a higher order law.
- **Third Law**: A robot must protect its own existence, as long as such protection does not conflict with a higher order law.

Advantages and Disadvantages of Robots

- Robotics and automation can, in many situations, increase product productivity, safety, efficiency, quality, and consistency.
- Robots can work in hazardous environments (such as radiation, darkness, hot and cold, ocean bottoms, space, etc.) without needing life support, comfort, or safety concerns.
- Robots need no environmental comfort like lighting, air conditioning, ventilation, and noise protection.
- Robots work continuously without tire fatigue or boredom. They do not get mad, do not have hangovers, and need no medical insurance or vacation.
- Robots have repeatable precision at all times unless something happens to them or they wear out.
- Robots can be much more accurate than humans.
- Robots and their accessories and sensors can have capabilities beyond those of humans.
- Robots can process multiple stimuli or tasks simultaneously. Humans can only process one active stimulus.
- Robots replace human workers, causing economic hardship, worker dissatisfaction and resentment, and the need for retraining the replaced workforce.
- Robots cannot respond in emergencies unless the situation is predicted and the system includes the response. Safety measures are needed to ensure that they do not injure operators and other machines that are working with them.
- Robots, although superior in certain senses, have limited capabilities in:
    - Cognition, creativity, decision-making, and understanding
    - Degrees of freedom and dexterity
    - Sensors and vision systems

    - Real-time response
- Robots are costly due to:
    - Initial cost of equipment and installation
    - Need for integration into the manufacturing processes
    - Need for peripherals
    - Need for training
    - Need for programming

Classification of Robots

- **Fixed-sequence robot**: A device that performs the successive stages of a task according to a predetermined, unchanging method that is hard to modify.
- **Playback robot**: A human operator performs a task manually by leading the robot, which records the motions for later playback. The robot repeats the same motions according to the recorded information.
- **Numerical-control robot**: The operator supplies the robot with a movement program rather than teaching it the task manually.
- **Intelligent robot**: A robot with the means to understand its environment and the ability to successfully complete a task despite changes in the surrounding conditions under which it is to be performed.

Robotic Components

- **Effectors** are the robot's appendages that move or move it, such as legs, arms, necks, and wrists. Effectors enable a robot to act on the environment. The major effectors in a ground robot are the wheels, tracks, or legs that give it navigational mobility. A new spate of work in robotics is centered on robot manipulators, essentially robot hands and arms.
- **Perception** is the set of sensors and sensing that provide a robot with the equivalent of eyes, ears, nose, smell, and touch. It generally requires a sensor, the device that collects a signal, and algorithms to interpret it. Perception is how a robot senses the environment.
- **Control** is analogous to the central nervous system, where a computer processor(s) provides the robot's inner and outer loop control. The control component contains the computations that allow an intelligent robot to maximize its chances of success.
- **Communication** is how a robot interacts with other agents if only the robot operator allows the exchange of information with other entities. Animals use bird and whale songs, displays of color and posture to communicate, while humans use natural language, gestures, proxemics, and other communication mechanisms.
- **Power** enables other functions; it duplicates the role of food and the digestive system in animals.

Robot System Elements

As a system, a robot consists of the following elements, which are integrated to form a whole.

- **Manipulator or rover**. The robot's main body consists of the links, joints, and other structural elements. Without other elements, the manipulator alone is not a robot.
- **End effector**. This part is connected to the last joint (hand) of a manipulator and generally handles objects, makes connections to other machines, or performs required tasks.
    - Generally, a robot's hand has provisions for attaching specialty end effectors specifically designed for a purpose. Welding torches, paint spray guns, glue-laying devices, and parts handlers are but a few examples.
- **Actuators**. Actuators are the “muscles” of the manipulators. The controller sends signals to the actuators, which move the robot’s joints and links.
    - Common types are servomotors, stepper motors, pneumatic actuators, and hydraulic actuators.
- **Sensors**. Sensors are used to collect information about the robot's internal state or communicate with the outside environment.
    - The robot controller needs to know where each link of the robot is to know the robot’s configuration.
    - Sensors are similar to a human major senses of sight, touch, hearing, taste, smell, and speech. While robots, on the other hand, are equipped with external sensory devices such as a vision system, touch

and tactile sensors, a speech synthesizer, and so on, that enable the robot to communicate with the outside world.

- **Controller**. The controller is similar to a cerebellum; although it does not have the brain's power, it still controls motions.
    - The controller receives its data from the processor (the brain of the system), controls the motions of the actuators, and coordinates the motions with the sensory feedback information.
- **Processor**. The processor is the brain of the robot. It calculates the motions of the robot's joints based on the programs it runs, determines how much and how fast each joint must move to achieve the desired location and speeds, and oversees the coordinated actions of the controller and the sensors.
- **Software**. Three groups of software programs are used in a robot.
    - (i) The operating system operates the processor.
    - (ii) Robotic software calculates the necessary motions of each joint based on the kinematic equations of the robot. This information is sent to the controller. This software may be at many levels, from machine language to sophisticated languages modern robots use.
    - (iii) A collection of application-oriented routines and programs are developed to use the robot or its peripherals for specific tasks, such as assembly, machine loading, material handling, etc.

References:

Cook, D. (2015). *Robot building for beginners* (3rd ed.). Apress Media LLC.
Kurfess, T. (2005). *Robotics and automation handbook*. CRC Press.
Murphy, R. (2019). *Introduction to AI robotics* (2nd ed.). MIT Press.
Niku, S. (2020). *Introduction to robotics: Analysis, control, applications* (3rd ed.). Wiley.